VIVIENNE HNIN

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EDUCATION

University of North Carolina at Chapel Hill — Chapel Hill, NC
B.A. in Computer Science, Minor in Neuroscience — *Expected May 2027*

- **Awards:** 1st Place, AfroPix '26, SolHacks '26, Quantum Hacks '26, 2nd Place, HackNC ·26
- **Coursework:** Algorithms, Data Structures, Databases, Computer Organization, Neural Networks

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, JavaScript, SQL
Security & Identity: Identity & Access Management (IAM) concepts, Least Privilege, RBAC, Incident Response, Secure Software Development, REST APIs
Systems & Infrastructure: Linux, Docker, AWS, GitHub, NVIDIA Jetson, Object-Oriented Programming, Data Structures & Algorithms
Performance & Telemetry: Latency and memory benchmarking, Python diagnostics, telemetry analysis
Databases & Frameworks: PostgreSQL, PostGIS, MongoDB, Firebase, FastAPI, React, HTML/CSS
AI / ML: PyTorch, Hugging Face, OpenCV, MediaPipe, Pandas, NumPy, LoRA
Certifications: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, Lean Six Sigma Green Belt

EXPERIENCE

IT Technician — Aug 2025 – Present
UNC Chapel Hill & Kenan-Flagler Business School — *Chapel Hill, NC*

- Managed **identity lifecycle** (provisioning, **access control**, and deprovisioning) and incident response across **7,900+ enterprise endpoints** using SolarWinds.
- Debugged enterprise identity and **authentication issues**, escalating security incidents and enforcing least-privilege account practices.

AI Engineering Intern — May 2026 – Aug 2026
Utilyst Inc. — *Chapel Hill, NC*

- Built and containerized backend inference services using **Python, FastAPI, REST APIs, and Docker**, delivering reproducible, isolated deployments across local and NVIDIA Jetson environments.
- Designed Python benchmarking pipelines to measure **latency, memory utilization**, and model performance across multiple deployment configurations on resource-constrained hardware.
- Developed geospatial application infrastructure using **PostgreSQL/PostGIS**, FastAPI, and React, enabling real-time analysis and visualization of municipal utility assets.
- Integrated and evaluated Phi-3 Mini, Gemma 2B, and Llama 3.2 3B within production-oriented software workflows on NVIDIA Jetson Orin Nano.

Product Management Extern — Nov 2025 – Jan 2026
BeReal — *Remote*

- Analyzed A/B tests and product telemetry across **50,000+ user interactions**, synthesizing behavioral funnels, retention data, and insights from 20+ user interviews to identify activation opportunities.

Data Analyst Intern — Jun 2024 – Aug 2024
Nawarat Autoparts — *Yangon, Burma*

- Engineered Python and SQL ETL pipelines processing **100,000+ inventory and sales records**, reducing reporting time by 80% and data errors by 30%.
- Built data-processing and forecasting workflows using Pandas and statistical analysis, reducing stock shortages by 25% and contributing to 239% channel sales growth.

PROJECTS

Dearest: Full-Stack Privacy & Moderation Platform | **React, TypeScript, FastAPI, Qdrant** — Live Demo

- Engineered a full-stack application with a React frontend and FastAPI backend, implementing an **asynchronous 8-stage pipeline** for content moderation and **privacy enforcement** ahead of semantic retrieval.
- Built hybrid vector retrieval using cosine search and **Qdrant ANN**, improving Recall@5 from 0.385 to 0.658, while integrating spaCy NER for sensitive-data detection and Celery-based background processing.

LiftAI: Real-Time Computer Vision Web Application — Aug 2026 – Present

- Engineered a privacy-first, browser-based application using React, TypeScript, and MediaPipe Pose, running real-time computer vision inference entirely client-side with zero backend calls or data transmission.
- Implemented real-time geometry calculations across 33 pose landmarks and optimized rendering to sustain **45+ FPS** with **<22ms latency** while generating live skeletal overlays.

LEADERSHIP & INVOLVEMENT

Vice President — Aug 2025 – Present
RAM Engineering — *Chapel Hill, NC*

- Lead cross-functional teams across avionics, embedded systems, telemetry, and software for autonomous VTOL aircraft.
- Direct ground-station software and flight-control integration across embedded, telemetry, and avionics systems.
- Managed $35,000+ in 2026 engineering funds, supporting project teams, equipment purchases, and technical initiatives.